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**SUBJECT: AWS CERTIFIED SOLUTIONS
ARCHITECT – ASSOCIATE
CERTIFICATE EXAM-3**

BATCH : B 303

AWS-DEVOPS



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QUESTION 1

Your company needs to deploy an application in the company AWS account. The application will reside on EC2 instances in an Auto Scaling Group fronted by an Application Load Balancer. The company has been using Elastic Beanstalk to deploy the application due to limited AWS experience within the organization. The application now needs upgrades and a small team of subcontractors have been hired to perform these upgrades. Which web service can be used to provide users that you authenticate with short-term security credentials that can control access to your AWS resources?

☐ IAM user accounts

☐ IAM Group

☐ AWS STS

☐ AWS SSO

✗ IAM user accounts

✗ IAM Group

✓ AWS STS

AWS Security Token Service (AWS STS) is the service that you can use to create and provide trusted users with temporary security credentials that can control access to your AWS resources. Temporary security credentials work almost identically to the long-term access key credentials that your IAM users can use. You can use the AWS Security Token Service (AWS STS) to create and provide trusted users with temporary security credentials that can control access to your AWS resources. Temporary security credentials work almost identically to the long-term access key credentials that your IAM users can use, with the following differences: Temporary security credentials are short-term, as the name implies. They can be configured to last for anywhere from a few minutes to several hours. After the credentials expire, AWS no longer recognizes them or allows any kind of access from API requests made with them. Temporary security credentials are not stored with the user but are generated dynamically and provided to the user when requested. When (or even before) the temporary security credentials expire, the user can request new credentials, as long as the user requesting them still has permissions to do so.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_temp.html

✗ AWS SSO

QUESTION 2

An insurance company is creating an application that will perform analytics in near real-time on huge datasets in the terabyte range, and potentially even petabyte. The company is evaluating an AWS data storage option. Which AWS service will allow storage of petabyte-scale data and also allow fast complex queries over a large number of rows?

☐ DynamoDB

☐ Redshift

☐ ElastiCache

☐ RDS

QUESTION 2

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✗ DynamoDB

✓ Redshift

Amazon Redshift is a fully-managed, petabyte-scale data warehouse service in the Cloud. An Amazon Redshift data warehouse is a collection of computing resources called nodes, which are organized into a group called a cluster. Each cluster runs an Amazon Redshift engine and contains one or more databases.

Reference: [Amazon Redshift](#)

<https://docs.aws.amazon.com/redshift/latest/gsg/getting-started.html>

✗ ElastiCache

✗ RDS

QUESTION 3

You have joined a newly formed software company as a Solutions Architect. It is a small company, and you are the only employee with AWS experience. The owner has asked for your recommendations to ensure that the AWS resources are deployed to proactively remain within budget. Which AWS service can you use to help ensure you don't have cost overruns for your AWS resources?

☐ Cost Explorer

☐ Billing and Cost Management

☐ AWS Budgets

☐ Inspector

- ✗ Cost Explorer
- ✗ Billing and Cost Management

✓ AWS Budgets

Correct - AWS Budgets gives you the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. And remember the keyword, proactively. With AWS Budgets, we can be proactive about attending to cost overruns before they become a major budget issue at the end of the month or quarter. Budgets can be tracked at the monthly, quarterly, or yearly level, and you can customize the start and end dates. You can further refine your budget to track costs associated with multiple dimensions, such as AWS service, linked account, tag, and others. Budget alerts can be sent via email and/or Amazon Simple Notification Service (SNS) topic. You can also use AWS Budgets to set a custom reservation utilization target and receive alerts when your utilization drops below the threshold you define. RI utilization alerts support Amazon EC2, Amazon RDS, Amazon Redshift, and Amazon ElastiCache reservations. Budgets can be created and tracked from the AWS Budgets dashboard, or via the Budgets API.

<https://aws.amazon.com/aws-cost-management/aws-budgets/>

- ✗ Inspector

QUESTION 4

Your company has performed a Disaster Recovery drill which failed to meet the Recovery Time Objective (RTO) desired by executive management. The failure was due in large part to the amount of time taken to restore proper functioning on the database side. You have given management a recommendation of implementing synchronous data replication for the RDS database to help meet the RTO. Which of these options can perform synchronous data replication in RDS?

☐ DAX

☐ Read replicas

☐ RDS Multi-AZ

☐ AWS Database Migration Service

- ✗ DAX
- ✗ Read replicas
- ✓ RDS Multi-AZ

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for RDS database (DB) instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ). Each AZ runs on its own physically distinct, independent infrastructure, and is engineered to be highly reliable. In case of an infrastructure failure, Amazon RDS performs an automatic failover to the standby (or to a read replica in the case of Amazon Aurora), so that you can resume database operations as soon as the failover is complete. Since the endpoint for your DB instance remains the same after a failover, your application can resume database operation without the need for manual administrative intervention. <https://aws.amazon.com/rds/features/multi-az/>

- ✗ AWS Database Migration Service

QUESTION 5

You have been evaluating the NACLs in your company. Currently, you are looking at the default network ACL. Which statement is true about NACLs?

- ☐ The default configuration of the default NACL is Deny, and the default configuration of a custom NACL is Allow.
- ☐ The default configuration of the default NACL is Allow, and the default configuration of a custom NACL is Allow.
- ☐ The default configuration of the default NACL is Allow, and the default configuration of a custom NACL is Deny.
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- ✗ The default configuration of the default NACL is Allow, and the default configuration of a custom NACL is Allow.
- ✓ The default configuration of the default NACL is Allow, and the default configuration of a custom NACL is Deny.

Your VPC automatically comes with a modifiable default network ACL. By default, it allows all inbound and outbound IPv4 traffic and, if applicable, IPv6 traffic. You can create a custom network ACL and associate it with a subnet. By default, each custom network ACL denies all inbound and outbound traffic until you add rules.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#default-network-acl>

- ✗ The default configuration of the default NACL is Deny, and the default configuration of a custom NACL is Deny.

QUESTION 6

You are designing a new application for a social media gaming company and they want to be able to use people's facebook account to login and to sign up to the new app. They've asked that you use Cognito to achieve this. However, management wants to know the steps involved for user authentication. Which of the steps below is the correct order to authenticate with Cognito?

- ☐ Step 1 - Access AWS services using credentials. Step 2 - Exchange tokens and get AWS credentials. Step 3 - Authenticate and get tokens.
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- ✓ Step 1 - Authenticate and get tokens. Step 2 - Exchange tokens and get AWS credentials. Step 3 - Access AWS services using credentials.

| This is how the authentication process works.

QUESTION 7

You have a typical architecture for an Application Load Balancer fronting an Auto Scaling group of EC2 instances, backed by an RDS MySQL database. Your Application Load Balancer is performing health checks on the EC2 instances. What actions will be taken if an instance fails these health checks?

- ☐ The ALB notifies the Auto Scaling group that the instance is down.
- ☐ The instance is replaced by the ALB.
- ☐ The ALB stops sending traffic to the instance.
- ☐ The instance is terminated by the ALB.

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✗ The instance is replaced by the ALB.

✓ The ALB stops sending traffic to the instance.

The load balancer routes requests only to the healthy instances. When the load balancer determines that an instance is unhealthy, it stops routing requests to that instance. The load balancer resumes routing requests to the instance when it has been restored to a healthy state. <https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/elb-healthchecks.html>

✗ The instance is terminated by the ALB.

QUESTION 8

You have taken over management of several instances in the company AWS environment. You want to quickly review scripts used to bootstrap the instances at runtime. A URL command can be used to do this. What can you append to the URL <http://169.254.169.254/latest/> to retrieve this data?

☐ user-data/

☐ instance-demographic-data/

☐ meta-data/

☐ instance-data/

QUESTION 8

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✓ ☒ `user-data/`

When you launch an instance in Amazon EC2, you have the option of passing user data to the instance that can be used to perform common automated configuration tasks and even run scripts after the instance starts. You can pass two types of user data to Amazon EC2: shell scripts and cloud-init directives.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instancedata-add-user-data.html> <https://docs.aws.amazon.com/AWSEC2/latest/WindowsGuide/ec2-instance-metadata.html> <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/user-data.html#user-data-shell-scripts>

✗ ☐ `instance-demographic-data/`

✗ ☐ `meta-data/`

✗ ☐ `instance-data/`

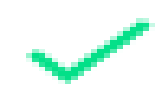
QUESTION 9

You have been evaluating the NACLs in your company. Most of the NACLs are configured the same:

```
100 All Traffic Allow  
200 All Traffic Deny  
* All Traffic Deny
```

If a request comes in, how will it be evaluated?

- ☐ The request will be allowed.
- ☐ All rules will be evaluated and the end result will be Deny.
- ☐ The default will deny traffic.
- ☐ The highest numbered rule will be used, a deny.



The request will be allowed.

Correct. Rules are evaluated starting with the lowest numbered rule. As soon as a rule matches traffic, it's applied immediately regardless of any higher-numbered rule that may contradict it. The following are the basic things that you need to know about network ACLs: Your VPC automatically comes with a modifiable default network ACL. By default, it allows all inbound and outbound IPv4 traffic and, if applicable, IPv6 traffic. You can create a custom network ACL and associate it with a subnet. By default, each custom network ACL denies all inbound and outbound traffic until you add rules. Each subnet in your VPC must be associated with a network ACL. If you don't explicitly associate a subnet with a network ACL, the subnet is automatically associated with the default network ACL. You can associate a network ACL with multiple subnets. However, a subnet can be associated with only one network ACL at a time. When you associate a network ACL with a subnet, the previous association is removed. A network ACL contains a numbered list of rules. We evaluate the rules in order, starting with the lowest-numbered rule, to determine whether traffic is allowed in or out of any subnet associated with the network ACL. The highest number that you can use for a rule is 32766. We recommend that you start by creating rules in increments (for example, increments of 10 or 100) so that you can insert new rules where you need to later on. A network ACL has separate inbound and outbound rules, and each rule can either allow or deny traffic. Network ACLs are stateless, which means that responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

QUESTION 10

The AWS team in a large company is spending a lot of time monitoring EC2 instances and maintenance when the instances report health check failures. How can you most efficiently automate this monitoring and repair?

- ☐ Create a Lambda function which can be triggered by a failed instance health check. Have the Lambda function deploy a CloudFormation template which can perform the creation of a new instance.
- ☐ Create an Amazon CloudWatch alarm that monitors an Amazon EC2 instance and automatically reboots the instance if a health check fails.
- ☐ Create a Lambda function which can be triggered by a failed instance health check. Have the Lambda function destroy the instance and spin up a new instance.
- ☐ Create a cron job which monitors the instances periodically and starts a new instance if a health check has failed.

- ✗ Create a Lambda function which can be triggered by a failed instance health check. Have the Lambda function deploy a CloudFormation template which can perform the creation of a new instance.
- ✓ Create an Amazon CloudWatch alarm that monitors an Amazon EC2 instance and automatically reboots the instance if a health check fails.

You can create an Amazon CloudWatch alarm that monitors an Amazon EC2 instance and automatically reboots the instance. The reboot alarm action is recommended for Instance Health Check failures (as opposed to the recover alarm action, which is suited for System Health Check failures). An instance reboot is equivalent to an operating system reboot. In most cases, it takes only a few minutes to reboot your instance. When you reboot an instance, it remains on the same physical host, so your instance keeps its public DNS name, private IP address, and any data on its instance store volumes. Rebooting an instance doesn't start a new instance billing hour, unlike stopping and restarting your instance.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/UsingAlarmActions.html>

- ✗ Create a Lambda function which can be triggered by a failed instance health check. Have the Lambda function destroy the instance and spin up a new instance.
- ✗ Create a cron job which monitors the instances periodically and starts a new instance if a health check has failed.

QUESTION 11

An Application Load Balancer is fronting an Auto Scaling Group of EC2 instances, and the instances are backed by an RDS database. The Auto Scaling Group has been configured to use the Default Termination Policy. You are testing the Auto Scaling Group and have triggered a scale-in. Which instance will be terminated first?

- ☐ The instance launched from the oldest launch configuration.
- ☐ The longest running instance.
- ☐ The Auto Scaling Group will randomly select an instance to terminate.
- ☐ The instance for which the load balancer stops sending traffic.

- ✓ The instance launched from the oldest launch configuration.

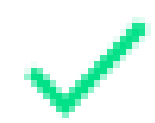
What do we know? The ASG is using the Default Termination Policy. The default termination policy is designed to help ensure that your instances span Availability Zones evenly for high availability. The default policy is kept generic and flexible to cover a range of scenarios. The default termination policy behavior is as follows: Determine which Availability Zones have the most instances, and at least one instance that is not protected from scale in. Determine which instances to terminate so as to align the remaining instances to the allocation strategy for the on-demand or spot instance that is terminating. This only applies to an Auto Scaling Group that specifies allocation strategies. For example, after your instances launch, you change the priority order of your preferred instance types. When a scale-in event occurs, Amazon EC2 Auto Scaling tries to gradually shift the on-demand instances away from instance types that are lower priority. Determine whether any of the instances use the oldest launch template or configuration: [For Auto Scaling Groups that use a launch template] Determine whether any of the instances use the oldest launch template unless there are instances that use a launch configuration. Amazon EC2 Auto Scaling terminates instances that use a launch configuration before instances that use a launch template. [For Auto Scaling Groups that use a launch configuration] Determine whether any of the instances use the oldest launch configuration. After applying all of the above criteria, if there are multiple unprotected instances to terminate, determine which instances are closest to the next billing hour. If there are multiple unprotected instances closest to the next billing hour, terminate one of these instances at random.

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-instance-termination.html>

QUESTION 12

After an IT Steering Committee meeting you have been put in charge of configuring a hybrid environment for the company's compute resources. You weigh the pros and cons of various technologies based on the requirements you are given. Your primary requirement is the necessity for a private, dedicated connection, which bypasses the Internet and can provide throughput of 10 Gbps. Which option will you select?

- ☐ VPC Peering
- ☐ AWS Direct Gateway
- ☐ AWS VPN
- ☐ AWS Direct Connect



AWS Direct Connect

Correct. AWS Direct Connect can reduce network costs, increase bandwidth throughput, and provide a more consistent network experience than internet-based connections. It uses industry-standard 802.1q VLANs to connect to Amazon VPC using private IP addresses. You can choose from an ecosystem of WAN service providers for integrating your AWS Direct Connect endpoint in an AWS Direct Connect location with your remote networks. AWS Direct Connect lets you establish 1 Gbps or 10 Gbps dedicated network connections (or multiple connections) between AWS networks and one of the AWS Direct Connect locations. You can also work with your provider to create sub-1G connection or use link aggregation group (LAG) to aggregate multiple 1 gigabit or 10 gigabit connections at a single AWS Direct Connect endpoint, allowing you to treat them as a single, managed connection. A Direct Connect gateway is a globally available resource to enable connections to multiple Amazon VPCs across different regions or AWS accounts. <https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-direct-connect.html>

QUESTION 13

You work for an Australian company, who are currently being audited and need some compliance reports regarding your applications that are hosted on AWS. Specifically, they need a Australian Hosting Certification Framework - Strategic Certification certificate. You need to get this as quickly as possible. What should you do?

- ☐ Use Amazon Detective to generate the report
- ☐ Use AWS Trusted Advisor to generate the report
- ☐ Use AWS Certificate Manager to generate the certificate
- ☐ Use AWS Artifact to download the certificate

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- ✗ Use Amazon Detective to generate the report
- ✗ Use AWS Trusted Advisor to generate the report
- ✗ Use AWS Certificate Manager to generate the certificate
- ✓ Use AWS Artifact to download the certificate

AWS Artifact is a single source you can visit to get the compliance-related information that matters to you, such as AWS security and compliance reports or select online agreements.

QUESTION 14

A small startup company has begun using AWS for all of its IT infrastructure. The company has one AWS Solutions Architect and the demands for their time are overwhelming. The software team has been given permission to deploy their Python and PHP applications on their own. They would like to deploy these applications without having to worry about the underlying infrastructure. Which AWS service would they use for deployments?

☐ CloudFront

☐ Elastic Beanstalk

☐ CloudFormation

☐ CodeDeploy

✗ CloudFront

✓ Elastic Beanstalk

With Elastic Beanstalk, you can quickly deploy and manage applications in the AWS Cloud without having to learn about the infrastructure that runs those applications. Elastic Beanstalk reduces management complexity without restricting choice or control. You simply upload your application, and Elastic Beanstalk automatically handles the details of capacity provisioning, load balancing, scaling, and application health monitoring. Elastic Beanstalk supports applications developed in Go, Java, .NET, Node.js, PHP, Python, and Ruby. When you deploy your application, Elastic Beanstalk builds the selected supported platform version and provisions one or more AWS resources, such as Amazon EC2 instances, to run your application.

<https://docs.aws.amazon.com/elasticbeanstalk/latest/dg/Welcome.html>

✗ CloudFormation

✗ CodeDeploy

QUESTION 15

Several S3 Buckets have been deleted and a few EC2 instances have been terminated. Which AWS service can you use to determine who took these actions?

☐ AWS Inspector

☐ AWS CloudTrail

☐ Trusted Advisor

☐ AWS CloudWatch

QUESTION 15

Several S3 Buckets have been deleted and a few EC2 instances have been terminated. Which AWS service can you use to determine who took these actions?

✗ AWS Inspector

✓ AWS CloudTrail

AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. With CloudTrail, you can log, continuously monitor, and retain account activity related to actions across your AWS infrastructure. CloudTrail provides event history of your AWS account activity, including actions taken through the AWS Management Console, AWS SDKs, command-line tools, and other AWS services. This event history simplifies security analysis, resource change tracking, and troubleshooting. In addition, you can use CloudTrail to detect unusual activity in your AWS accounts. These capabilities help simplify operational analysis and troubleshooting.

✗ Trusted Advisor

✗ AWS CloudWatch

QUESTION 16

A new startup is considering the advantages of using Amazon DynamoDB versus a traditional relational database in AWS RDS. The NoSQL nature of DynamoDB presents a small learning curve to the team members who all have experience with traditional databases. The company will have multiple databases, and the decision will be made on a case-by-case basis. Which of the following use cases would favor Amazon DynamoDB?

(Choose 3)

- ☐ Storing metadata for S3 objects
- ☐ Online analytical processing (OLAP)/data warehouse implementations
- ☐ High-performance reads and writes for online transaction processing (OLTP) workloads
- ☐ Managing web session data
- ☐ Storing binary large object (BLOB) data
- ☐ Strong referential integrity between tables

✓ Storing metadata for S3 objects

Storing metadata for Amazon S3 objects is correct because the Amazon DynamoDB stores structured data indexed by primary key and allows low-latency read and write access to items ranging from 1 byte up to 400KB. Amazon S3 stores unstructured blobs and is suited for storing large objects up to 5 TB. In order to optimize your costs across AWS services, large objects or infrequently accessed data sets should be stored in Amazon S3, while smaller data elements or file pointers (possibly to Amazon S3 objects) are best saved in Amazon DynamoDB.

✗ Online analytical processing (OLAP)/data warehouse implementations

✓ High-performance reads and writes for online transaction processing (OLTP) workloads

High-performance reads and writes are easy to manage with Amazon DynamoDB, and you can expect performance that is effectively constant across widely varying loads. [How to determine if Amazon DynamoDB is appropriate for your needs, and then plan your migration.](#)

✓ Managing web session data

Amazon DynamoDB is a NoSQL database that supports key-value and document data structures. A key-value store is a database service that provides support for storing, querying, and updating collections of objects that are identified using a key and values that contain the actual content being stored. Meanwhile, a document data store provides support for storing, querying, and updating items in a document format such as JSON, XML, and HTML. Amazon DynamoDB's fast and predictable performance characteristics make it a great match for handling session data. Plus, since it's a fully-managed NoSQL database service, you avoid all the work of maintaining and operating a separate session store. [Amazon DynamoDB Session Manager for Apache Tomcat.](#)

✗ Storing binary large object (BLOB) data

✗ Strong referential integrity between tables

QUESTION 17

A company is going to use several EC2 instances to host various reference applications. The applications are expected to receive steady and relatively low traffic. These applications are expected to run for 3 years, at which time the applications will be evaluated for upgrade. What type of EC2 will meet this requirement considering cost as an additional factor?

☐ Dedicated Hosts

☐ Spot

☐ Reserved

☐ On-Demand

✕ Dedicated Hosts

✕ Spot

✓ Reserved

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them. For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

<https://aws.amazon.com/ec2/pricing/> <https://aws.amazon.com/ec2/pricing/reserved-instances/>

✕ On-Demand

QUESTION 18

You are evaluating the security setting within the main company VPC. There are several NACLs and security groups to evaluate and possibly edit. What is true regarding NACLs and security groups?

- ☐ Network ACLs are stateful, and security groups are stateless.
- ☐ Network ACLs and security groups are both stateful.
- ☐ Network ACLs are stateless, and security groups are stateful.
- ☐ Network ACLs and security groups are both stateless.

✓ Network ACLs are stateless, and security groups are stateful.

Network ACLs are stateless, which means that responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#default-network-acl> The following are the basic characteristics of security groups for your VPC:

- There are quotas on the number of security groups that you can create per VPC, the number of rules that you can add to each security group, and the number of security groups that you can associate with a network interface. For more information, see Amazon VPC quotas.
- You can specify allow rules, but not deny rules.
- You can specify separate rules for inbound and outbound traffic.
- When you create a security group, it has no inbound rules. Therefore, no inbound traffic originating from another host to your instance is allowed until you add inbound rules to the security group.
- By default, a security group includes an outbound rule that allows all outbound traffic. You can remove the rule and add outbound rules that allow specific outbound traffic only. If your security group has no outbound rules, no outbound traffic originating from your instance is allowed.
- Security groups are stateful. If you send a request from your instance, the response traffic for that request is allowed to flow in regardless of inbound security group rules. Responses to allowed inbound traffic are allowed to flow out, regardless of outbound rules.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html

QUESTION 19

A financial tech company has decided to begin migrating their applications to the AWS cloud. Currently, they host their entire application using several self-managed Kubernetes clusters. One of their major concerns during this migration is monitoring and collecting system metrics due to the very large-scale deployments that are in place. Your Chief Technology Officer wants to avoid using AWS-proprietary technology-based monitoring services and instead leverage existing, well-known, open-source applications to help meet the monitoring requirements. Which combination of the following AWS services would best fit the company requirements while minimizing operational overhead?

(Choose 2)

- ☐ AWS Config
- ☐ Prometheus on Auto Scaling EC2 Instances
- ☐ Grafana on Auto Scaling EC2 Instances
- ☐ AWS Managed Grafana
- ☐ AWS Managed Service for Prometheus

- ✗ AWS Config
- ✗ Prometheus on Auto Scaling EC2 Instances
- ✗ Grafana on Auto Scaling EC2 Instances

- ✓ AWS Managed Grafana

Grafana is a well-known open-source analytics and monitoring application. Amazon Managed Grafana offers a fully managed service for infrastructure for data visualizations. You can leverage this service to query, correlate, and visualize operational metrics from multiple sources. Reference: [What is Amazon Managed Grafana](#)

- ✓ AWS Managed Service for Prometheus

Prometheus offers open-source monitoring. Amazon Managed Service for Prometheus is a serverless, Prometheus-compatible monitoring service for container metrics. It is perfect for monitoring Kubernetes clusters at scale. Reference: [What is Amazon Managed Prometheus](#)

QUESTION 20

A new startup company decides to use AWS to host their web application. They configure a VPC as well as two subnets within the VPC. They also attach an internet gateway to the VPC. In the first subnet, they create the EC2 instance which will host their web application. They finish the configuration by making the application accessible from the Internet. The second subnet hosts their database and they don't want the database accessible from the Internet. Which statement best describes this scenario?

- ☐ The web server is in a public subnet, and the database server is in a public subnet. The public subnet has a route to the internet gateway in the route table.
- ☐ The web server is in a public subnet, and the database server is in a private subnet. The public subnet has a route to the internet gateway in the route table.
- ☐ The web server is in a private subnet, and the database server is in a public subnet. The public subnet has a route to the internet gateway in the route table.
- ☐ The web server is in a private subnet, and the database server is in a private subnet. A third subnet has a route to the Internet Gateway, which allows internet access.

- ✓ The web server is in a public subnet, and the database server is in a private subnet. The public subnet has a route to the internet gateway in the route table.

An internet gateway is a horizontally-scaled, redundant, and highly available VPC component that allows communication between your VPC and the Internet. An internet gateway serves two purposes: to provide a target in your VPC route tables for internet-routable traffic, and to perform network address translation (NAT) for instances that have been assigned public IPv4 addresses. An internet gateway supports IPv4 and IPv6 traffic. It does not cause availability risks or bandwidth constraints on your network traffic. To enable access to or from the Internet for instances in a subnet in a VPC, you must do the following:

- Attach an internet gateway to your VPC.
- Add a route to your subnet's route table that directs internet-bound traffic to the internet gateway. If a subnet is associated with a route table that has a route to an internet gateway, it's known as a public subnet. If a subnet is associated with a route table that does not have a route to an internet gateway, it's known as a private subnet.
- Ensure that instances in your subnet have a globally-unique IP address (public IPv4 address, Elastic IP address, or IPv6 address).
- Ensure that your network access control lists and security group rules allow the relevant traffic to flow to and from your instance.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_Internet_Gateway.html

QUESTION 21

Your company needs to shift an application to the cloud. You are looking for a solution to collect, process, gain immediate insight, and then transfer the application data to AWS. Part of this effort also includes moving a large data warehouse into AWS. The warehouse is 50TB, and it would take over a month to migrate the data using the current bandwidth available. What is the best option available to perform this one time migration considering both cost and performance aspects?

☐ AWS Snowball Edge

☐ AWS VPN

☐ AWS SnowMobile

☐ AWS Direct Connect

✓ AWS Snowball Edge

The AWS Snowball Edge is a type of Snowball device with on-board storage and compute power for select AWS capabilities. Snowball Edge can undertake local processing and edge-computing workloads in addition to transferring data between your local environment and the AWS Cloud.

Each Snowball Edge device can transport data at speeds faster than the internet. This transport is done by shipping the data in the appliances through a regional carrier. The appliances are rugged shipping containers, complete with E Ink shipping labels. The AWS Snowball Edge device differs from the standard Snowball because it can bring the power of the AWS Cloud to your on-premises location, with local storage and compute functionality.

Snowball Edge devices have three options for device configurations: storage optimized, compute optimized, and with GPU. When this guide refers to Snowball Edge devices, it's referring to all options of the device. Whenever specific information applies to only one or more optional configurations of devices, like how the Snowball Edge with GPU has an on-board GPU, it will be called out. For more information, see Snowball Edge Device Options. <https://docs.aws.amazon.com/snowball/latest/developer-guide/whatisedge.html> <https://docs.aws.amazon.com/snowball/latest/ug/device-differences.html>

QUESTION 22

A consultant is hired by a small company to configure an AWS environment. The consultant begins working with the VPC and launching EC2 instances within the VPC. The initial instances will be placed in a public subnet. The consultant begins to create security groups. The consultant has launched several instances, created security groups, and has associated security groups with instances. The consultant wants to change the security groups that are associated with the instance. Which statement is true?

- ☐ You can't change the security groups for an instance when the instance is in the running or stopped state.
- ☐ You can't change security groups. Create a new instance and attach the desired security groups.
- ☐ You can change the security groups for an instance when the instance is in the running or stopped state.
- ☐ You can change the security groups for an instance when the instance is in the pending or stopped state.

- ✗ You can't change the security groups for an instance when the instance is in the running or stopped state.
- ✗ You can't change security groups. Create a new instance and attach the desired security groups.
- ✓ You can change the security groups for an instance when the instance is in the running or stopped state.

Correct. After you launch an instance into a VPC, you can change the security groups that are associated with the instance. You can change the security groups for an instance when the instance is in the running or stopped state.

https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html

- ✗ You can change the security groups for an instance when the instance is in the pending or stopped state.

QUESTION 23

You work for a Defense contracting company. The company develops software applications which perform intensive calculations in the area of Mechanical Engineering related to metals for ship building. You have a 3-year contract and decide to purchase reserved EC2 instances for a 3-year duration. You are informed that the particular program has been cancelled abruptly and negotiations have brought the contract to an amicable conclusion one year early. What can you do to stop incurring charges and save money on the EC2 instances?

- ☐ Sell the reserved instances on the Reserved Instance Marketplace.
- ☐ Convert the instances to Spot Instances and allow them to go away through attrition.
- ☐ Change the instance states from running to stopped.
- ☐ Write AWS and ask to terminate the contract.

- ✓ Sell the reserved instances on the Reserved Instance Marketplace.

The Reserved Instance Marketplace is a platform that supports the sale of third-party and AWS customers' unused Standard Reserved Instances, which vary in term lengths and pricing options. For example, you may want to sell Reserved Instances after moving instances to a new AWS Region, changing to a new instance type, ending projects before the term expiration, when your business needs change, or if you have unneeded capacity.

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ri-market-general.html>

- ✗ Convert the instances to Spot Instances and allow them to go away through attrition.
- ✗ Change the instance states from running to stopped.
- ✗ Write AWS and ask to terminate the contract.

QUESTION 24

A company has created a mobile application that is hugely popular. The initial plan was to give each user login credentials to the application. But due to the volume of users, this idea has become impractical. What service can you use to allow outside users to login through a third party such as Facebook, Amazon, Google or Apple?

☐ Amazon Cognito

☐ AWS IAM

☐ Google Authenticator

☐ AWS cross account access

✓ Amazon Cognito

Correct - Amazon Cognito provides authentication, authorization, and user management for your web and mobile apps. Your users can sign in directly with a user name and password or through a third party such as Facebook, Amazon, Google, or Apple. The two main components of Amazon Cognito are user pools and identity pools. User pools are user directories that provide sign-up and sign-in options for your app users. Identity pools enable you to grant your users access to other AWS services. You can use identity pools and user pools separately or together.

<https://docs.aws.amazon.com/cognito/latest/developerguide/what-is-amazon-cognito.html>

✗ AWS IAM

✗ Google Authenticator

✗ AWS cross account access

QUESTION 25

You have been given an assignment to configure Network ACLs in your VPC. Before configuring the NACLs, you need to understand how the NACLs are evaluated. How are NACL rules evaluated?

- ☐ NACL rules are evaluated by rule number from highest to lowest, and executed immediately when a matching rule is found.
- ☐ All NACL rules that you configure are evaluated before traffic is passed through.
- ☐ NACL rules are evaluated by rule number from highest to lowest, and all are evaluated before traffic is passed through.
- ☐ NACL rules are evaluated by rule number from lowest to highest and executed immediately when a matching rule is found.

- ✓ NACL rules are evaluated by rule number from lowest to highest and executed immediately when a matching rule is found.

Correct. You can add or remove rules from the default network ACL, or create additional network ACLs for your VPC. When you add or remove rules from a network ACL, the changes are automatically applied to the subnets that it's associated with. A network access control list (ACL) is an optional layer of security for your VPC that acts as a firewall for controlling traffic in and out of one or more subnets. You might set up network ACLs with rules similar to your security groups in order to add an additional layer of security to your VPC. The following are the parts of a network ACL rule:

- Rule number. Rules are evaluated starting with the lowest-numbered rule. As soon as a rule matches traffic, it's applied regardless of any higher-numbered rule that might contradict it.
- Type. The type of traffic, for example, SSH. You can also specify all traffic or a custom range.
- Protocol. You can specify any protocol that has a standard protocol number. For more information, see Protocol Numbers. If you specify ICMP as the protocol, you can specify any or all of the ICMP types and codes.
- Port range. The listening port or port range for the traffic. For example, 80 for HTTP traffic.
- Source. [Inbound rules only] The source of the traffic (CIDR range).
- Destination. [Outbound rules only] The destination for the traffic (CIDR range).
- Allow/Deny. Whether to allow or deny the specified traffic.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html#nacl-rules>

QUESTION 26

You are working for a startup company with a small number of employees. The company expects rapid growth and you have been assigned to configure existing users and onboard new users with IAM privileges and logins. You intend to create IAM groups for the company departments and add new users to the appropriate group when you onboard them. You begin creating policies to assign permissions and attach them to the appropriate group. What is the best practice when giving users permissions in IAM policies?

- ☐ Use the principle of least privilege.
- ☐ Create a policy for each department head granting root access.
- ☐ Use the principle of top-down privilege.
- ☐ Grant all permissions to each AWS service the user will work with.

✓ Use the principle of least privilege.

When you create IAM policies, follow the standard security advice of granting least privilege, or granting only the permissions required to perform a task. Determine what users (and roles) need to do and then craft policies that allow them to perform only those tasks. Start with a minimum set of permissions and grant additional permissions as necessary. Doing so is more secure than starting with permissions that are too lenient and then trying to tighten them later.

<https://docs.aws.amazon.com/IAM/latest/UserGuide/best-practices.html#use-groups-for-permissions>

✗ Create a policy for each department head granting root access.

✗ Use the principle of top-down privilege.

✗ Grant all permissions to each AWS service the user will work with.

QUESTION 27

You have been hired as a Solutions Architect for a company that pairs photos with related story narratives in PDF format. The company needs to be able to store files in several different formats, such as PDF, JPG, PNG, Word, and several others. This storage needs to be highly durable. Which storage type will best meet this requirement?

☐ DynamoDB

☐ Amazon RDS

☐ EC2 instance store

☐ S3

- ✗ DynamoDB
- ✗ Amazon RDS
- ✗ EC2 instance store

✓ S3

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. This means customers of all sizes and industries can use it to store and protect any amount of data for a range of use cases, such as websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics. Amazon S3 provides easy-to-use management features so you can organize your data and configure finely-tuned access controls to meet your specific business, organizational, and compliance requirements. Amazon S3 is designed for 99.999999999% (11 9s) of durability, and stores data for millions of applications for companies all around the world.

<https://aws.amazon.com/s3/>

QUESTION 28

You have just started working at a company that is migrating from a physical data center into AWS. Currently, you have 25 TB of data that needs to be moved to an S3 bucket. Your company has just finished setting up a 1 GB Direct Connect drop, but you will not have a VPN up and running for 30 days. This data needs to be encrypted during transit and at rest and must be uploaded to the S3 bucket within 21 days. What is the best way to meet these requirements?

- ☐ Order a Snowcone device to transmit the data.
- ☐ Upload the data to S3 using your public internet connection.
- ☐ Upload the data using Direct Connect.
- ☐ Use a Snowball device to transmit the data.

QUESTION 28

You have just started working at a company that is migrating from a physical data center into AWS. Currently, you have 25 TB of data that needs to be moved to an S3 bucket. Your company has just finished setting up a 1 GB Direct Connect drop, but you will not have a VPN up and running for 30 days. This data needs to be encrypted during transit and at rest and must be uploaded to the S3 bucket within 21 days. What is the best way to meet these requirements?

- ✗ Order a Snowcone device to transmit the data.
- ✗ Upload the data to S3 using your public internet connection.
- ✗ Upload the data using Direct Connect.
- ✓ Use a Snowball device to transmit the data.

This would be the perfect choice to transmit your data. Snowball encrypts your data, so all the security and speed requirements would be met. Reference: [Encryption in AWS Snowball](#)

QUESTION 29

A team of architects is designing a new AWS environment for a company which wants to migrate to the Cloud. The architects are considering the use of EC2 instances with instance store volumes. The architects realize that the data on the instance store volumes are ephemeral. Which action will not cause the data to be deleted on an instance store volume?

- ☐ Hardware disk failure.
- ☐ The underlying disk drive fails.
- ☐ Reboot
- ☐ Instance is stopped

- ✗ Hardware disk failure.
- ✗ The underlying disk drive fails.

✓ Reboot

Correct. Some Amazon Elastic Compute Cloud (Amazon EC2) instance types come with a form of directly attached, block-device storage known as the instance store. The instance store is ideal for temporary storage, because data in the instance store is lost under any of the following circumstances:

- The underlying disk drive fails
- The instance stops
- The instance terminates
- Instance is terminated
- Hardware disk failure

References: <https://aws.amazon.com/premiumsupport/knowledge-center/instance-store-vs-ebs/>

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html>

- ✗ Instance is stopped

QUESTION 30

Recently, you've been experiencing issues with your dynamic application that is running on EC2 instances. These instances aren't able to keep up with the amount of traffic being sent to them, and customers are getting timeouts. Upon further investigation, there is no discernible traffic pattern for these surges. The application can be easily containerized. What can you do to fix the problem while keeping cost in mind?

- ☐ Migrate the web application to S3. Enable static website hosting.
- ☐ Increase the minimum and maximum count on the EC2 Auto Scaling group.
- ☐ Migrate the application to ECS. Use Fargate to run the required tasks.
- ☐ Create a second Auto Scaling group of EC2 instances. When the first is overwhelmed, post the overflow traffic to an SQS queue.

- ✗ Migrate the web application to S3. Enable static website hosting.
- ✗ Increase the minimum and maximum count on the EC2 Auto Scaling group.
- ✓ Migrate the application to ECS. Use Fargate to run the required tasks.
 - This would be a perfect use case for Fargate, as the workload is unpredictable. It will automatically scale in and out based on the workload being thrown at it.
<https://aws.amazon.com/fargate/>
- ✗ Create a second Auto Scaling group of EC2 instances. When the first is overwhelmed, post the overflow traffic to an SQS queue.

QUESTION 31

After an IT Steering Committee meeting, you have been put in charge of configuring a hybrid environment for the company's compute resources. You weigh the pros and cons of various technologies, such as VPN and Direct Connect, and based on the requirements you have decided to configure a VPN connection. What features and advantages can a VPN connection provide?

- ☐ It provides a private, dedicated network connection between an on-premises network and the VPC.
- ☐ It provides a cost-effective, private network connection that bypasses the internet.
- ☐ It provides a connection between an on-premises network and a VPC, using a secure and private connection with IPsec and TLS.
- ☐ It provides a network connection between two VPCs that can route traffic using IPv4 or IPv6.

- ✓ It provides a connection between an on-premises network and a VPC, using a secure and private connection with IPsec and TLS.

Correct: A VPC/VPN Connection utilizes IPsec to establish encrypted network connectivity between your intranet and Amazon VPC over the Internet. VPN Connections can be configured in minutes and are a good solution if you have an immediate need, have low-to-modest bandwidth requirements, and can tolerate the inherent variability in Internet-based connectivity.

AWS Client VPN is a managed client-based VPN service that enables you to securely access your AWS resources or your on-premises network. With AWS Client VPN, you configure an endpoint to which your users can connect to establish a secure TLS VPN session. This enables clients to access resources in AWS or on-premises from any location using an OpenVPN-based VPN client.

You can create an IPsec VPN connection between your VPC and your remote network. On the AWS side of the Site-to-Site VPN connection, a virtual private gateway or transit gateway provides two VPN endpoints (tunnels) for automatic failover. You configure your customer gateway device on the remote side of the Site-to-Site VPN connection.

<https://docs.aws.amazon.com/vpc/latest/userguide/vpn-connections.html>

QUESTION 32

A data company has implemented a subscription service for storing video files. There are two levels of subscription: personal and professional use. The personal users can upload a total of 5 GB of data, and professional users can upload as much as 5 TB of data. The application can upload files of size up to 1 TB to an S3 Bucket. What is the best way to upload files of this size?

- ☐ Multipart upload
- ☐ Single-part Upload
- ☐ AWS Snowball
- ☐ AWS SnowMobile

✓ Multipart upload

The Multipart upload API enables you to upload large objects in parts. You can use this API to upload new large objects or make a copy of an existing object (see Operations on Objects). Multipart uploading is a three-step process: You initiate the upload, you upload the object parts, and after you have uploaded all the parts, you complete the multipart upload. Upon receiving the complete multipart upload request, Amazon S3 constructs the object from the uploaded parts, and you can then access the object just as you would any other object in your bucket. You can list all of your in-progress multipart uploads or get a list of the parts that you have uploaded for a specific multipart upload. Each of these operations is explained in this section.

<https://docs.aws.amazon.com/AmazonS3/latest/dev/mpuoverview.html>

✗ Single-part Upload

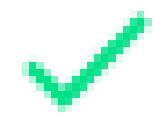
✗ AWS Snowball

✗ AWS SnowMobile

QUESTION 33

You are working for a large financial institution and preparing for disaster recovery and upcoming DR drills. A key component in the DR plan will be the database instances and their data. An aggressive Recovery Time Objective (RTO) dictates that the database needs to be synchronously replicated. Which configuration can meet this requirement?

- ☐ RDS Multi-Region
- ☐ RDS read replicas
- ☐ AWS Lambda triggers a CloudFormation template launch in another Region.
- ☐ RDS Multi-AZ



RDS Multi-AZ

Amazon RDS Multi-AZ deployments provide enhanced availability and durability for RDS database (DB) instances, making them a natural fit for production database workloads. When you provision a Multi-AZ DB instance, Amazon RDS automatically creates a primary DB Instance and synchronously replicates the data to a standby instance in a different Availability Zone (AZ). Each AZ runs on its own physically distinct, independent infrastructure, and is engineered to be highly reliable. In case of an infrastructure failure, Amazon RDS performs an automatic failover to the standby (or to a read replica in the case of Amazon Aurora), so that you can resume database operations as soon as the failover is complete. Since the endpoint for your DB instance remains the same after a failover, your application can resume database operation without the need for manual administrative intervention.

<https://aws.amazon.com/rds/features/multi-az/>

QUESTION 34

You are designing an architecture that will house an Auto Scaling group of EC2 instances. The application hosted on the instances is expected to be extremely popular. Forecasts for traffic to this site predict very high traffic, and you will need a load balancer to handle tens of millions of requests per second while maintaining high throughput at ultra-low latency. You need to select the correct type of load balancer to front your Auto Scaling group to meet this high traffic requirement. Which load balancer should you select?

- ☐ You will need an Application Load Balancer to meet this requirement.
- ☐ You will need a Classic Load Balancer to meet this requirement.
- ☐ You will need a Network Load Balancer to meet this requirement.
- ☐ All the AWS load balancers meet the requirement and perform the same.

✓ You will need a Network Load Balancer to meet this requirement.

If extreme performance is needed for your application, AWS recommends that you use a Network Load Balancer. A Network Load Balancer operates at the connection level (Layer 4), routing connections to targets (Amazon EC2 instances, microservices, and containers) within Amazon VPC, based on IP protocol data. It is ideal for load balancing both TCP and UDP traffic, and is capable of handling millions of requests per second while maintaining ultra-low latencies. It is optimized to handle sudden and volatile traffic patterns while using a single static IP address per Availability Zone, and is integrated with other popular AWS services such as Auto Scaling, Amazon EC2 Container Service (ECS), Amazon CloudFormation, and AWS Certificate Manager (ACM).

References: [Network Load Balancer](#)

[Elastic Load Balancing Features](#)

QUESTION 35

A testing team is using a group of EC2 instances to run batch, automated tests on an application. The tests run overnight, but don't take all night. The instances sit idle for long periods of time and accrue unnecessary charges. What can you do to stop these instances when they are idle for long periods?

- ☐ Write a cron job which queries the instance status. If a certain status is met, have the cron job kick off CloudFormation to terminate the existing instance, and create a new instance from a template.
- ☐ You can create a CloudWatch alarm that is triggered when the average CPU utilization percentage has been lower than 10 percent for 4 hours, and stops the instance.
- ☐ Write a Python script which queries the instance status. Also write a Lambda function which can be triggered upon a certain status and stop the instance.
- ☐ Write a cron job which queries the instance status. Also write a Lambda function which can be triggered upon a certain status and stop the instance.

- ✓ You can create a CloudWatch alarm that is triggered when the average CPU utilization percentage has been lower than 10 percent for 4 hours, and stops the instance.

Adding Stop Actions to Amazon CloudWatch Alarms: You can create an alarm that stops an Amazon EC2 instance when a certain threshold has been met. For example, you may run development or test instances and occasionally forget to shut them off. You can create an alarm that is triggered when the average CPU utilization percentage has been lower than 10 percent for 24 hours, signaling that it is idle and no longer in use. You can adjust the threshold, duration, and period to suit your needs, plus you can add an SNS notification, so that you will receive an email when the alarm is triggered.

Amazon EC2 instances that use an Amazon Elastic Block Store volume as the root device can be stopped or terminated, whereas instances that use the instance store as the root device can only be terminated.

<https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/UsingAlarmActions.html>

QUESTION 36

After being assigned to oversee the data storage within your organization, you begin looking at the monthly billing for S3. You notice that large amounts of data are sitting in S3, and after discussions with team members you find that a large amount of the data is historical data that needs to be kept for audit purposes. You detail the cost savings and get approval to move this data to Amazon Glacier for long-term storage. For what types of data is Glacier best suited?

(Choose 2)

☐ Cached data

☐ Infrequently accessed data

☐ Archival data

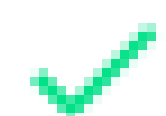
☐ Relational table data

✓ Infrequently accessed data

Correct. One of the purposes of Glacier is to store infrequently accessed data. It is important to understand the distinction between S3 IA and Glacier. S3 IA also can be used to store infrequently accessed data, but with S3 IA, you can retrieve the data immediately. With Glacier, data retrieval ranges from minutes to hours. So, if you do not need to retrieve infrequently accessed data immediately, then Glacier is a good choice and will provide cost savings. In transitioning S3 standard to Glacier you need to tell S3 which objects are to be archived to the new Glacier storage option, and under what conditions. You do this by setting up a lifecycle rule using the following elements:

- A prefix to specify which objects in the bucket are subject to the policy.
- A relative or absolute time specifier and a time period for transitioning objects to Glacier. The time periods are interpreted with respect to the object's creation date. They can be relative (migrate items that are older than a certain number of days) or absolute (migrate items on a specific date). An object age at which the object will be deleted from S3. This is measured from the original PUT of the object into the service, and the clock is not reset by a transition to Glacier. You can create a lifecycle rule in the AWS Management Console.

<https://aws.amazon.com/glacier/faqs/>



Archival data

Correct. Amazon S3 Glacier and S3 Glacier Deep Archive are a secure, durable, and extremely low-cost Amazon S3 cloud storage classes for data archiving and long-term backup. They are designed to deliver 99.999999999% durability, and provide comprehensive security and compliance capabilities that can help meet even the most stringent regulatory requirements. Customers can store data for as little as \$1 per terabyte per month, a significant savings compared to on-premises solutions. To keep costs low yet suitable for varying retrieval needs, Amazon S3 Glacier provides three options for access to archives, ranging from a few minutes to several hours. S3 Glacier Deep Archive provides two access options ranging from 12 to 48 hours.

<https://aws.amazon.com/glacier/>

QUESTION 37

An organization of about 100 employees has performed the initial setup of users in IAM. All users except administrators have the same basic privileges. But now it has been determined that 50 employees will have extra restrictions on EC2. They will be unable to launch new instances or alter the state of existing instances. What will be the quickest way to implement these restrictions?

- ☐ Create the appropriate policy. Place the restricted users in the new policy.
- ☐ Create the appropriate policy. With only 20 users, attach the policy to each user.
- ☐ Create the appropriate policy. Create a new group for the restricted users. Place the restricted users in the new group and attach the policy to the group.
- ☐ Create an IAM Role for the restrictions. Attach it to the EC2 instances.

- ✓ Create the appropriate policy. Create a new group for the restricted users. Place the restricted users in the new group and attach the policy to the group.

You manage access in AWS by creating policies and attaching them to IAM identities (users, groups of users, or roles) or AWS resources. A policy is an object in AWS that, when associated with an identity or resource, defines their permissions. AWS evaluates these policies when an IAM principal (user or role) makes a request. Permissions in the policies determine whether the request is allowed or denied. Most policies are stored in AWS as JSON documents. AWS supports six types of policies: identity-based policies, resource-based policies, permissions boundaries, Organizations SCPs, ACLs, and session policies. IAM policies define permissions for an action regardless of the method that you use to perform the operation. For example, if a policy allows the GetUser action, then a user with that policy can get user information from the AWS Management Console, the AWS CLI, or the AWS API. When you create an IAM user, you can choose to allow console or programmatic access. If console access is allowed, the IAM user can sign in to the console using a user name and password. Or if programmatic access is allowed, the user can use access keys to work with the CLI or API.

https://docs.aws.amazon.com/IAM/latest/UserGuide/access_policies.html

QUESTION 38

After an IT Steering Committee meeting, you have been put in charge of configuring a hybrid environment for the company's compute resources. You weigh the pros and cons of various technologies based on the requirements you are given. The decision you make is to go with Direct Connect. Which option best describes the features Direct Connect provides?

- ☐ A cost-effective, private network connection that bypasses the internet
- ☐ A connection between on-premises and VPC, using secure and private connection with IPsec and TLS
- ☐ A private, dedicated network connection between your facilities and AWS
- ☐ A network connection between two VPCs that can route traffic using IPv4 or IPv6

- ✓ A private, dedicated network connection between your facilities and AWS

Correct: AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your datacenter, office, or colocation environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than internet-based connections.

AWS Direct Connect makes it easy to establish a dedicated connection from an on-premises network to one or more VPCs in the same region. Using private VIF on AWS Direct Connect, you can establish private connectivity between AWS and your data center, office, or colocation environment.

AWS Direct Connect can reduce network costs, increase bandwidth throughput, and provide a more consistent network experience than internet-based connections.

<https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/aws-direct-connect.html>

<https://docs.aws.amazon.com/whitepapers/latest/aws-vpc-connectivity-options/network-to-amazon-vpc-connectivity-options.html>

QUESTION 39

Your company has a multi-account AWS environment with over 100 accounts. Each account belongs to a specific application team within the company, and they all fall within the same consolidated billing family. The company has just received funding for the next two years, but they are unsure about anything beyond that. With this in mind, they plan on aggressively deploying applications to AWS during the two years.

Recently, there was a massive spike in unplanned Amazon EC2 and AWS Lambda costs, causing significant financial stress.

What can an organization administrator do to maximize savings for the entire organization for this first year?

- ☐ Purchase a three-year All Upfront Compute Savings Plan.
- ☐ Purchase a three-year All Upfront EC2 Instance Savings Plan.
- ☐ Purchase a one-year All Upfront Compute Savings Plan.
- ☐ Purchase a one-year All Upfront EC2 Instance Savings Plan.

✗ Purchase a three-year All Upfront Compute Savings Plan.

✗ Purchase a three-year All Upfront EC2 Instance Savings Plan.

✓ Purchase a one-year All Upfront Compute Savings Plan.

This type of Savings Plan covers both Amazon EC2 and AWS Lambda function compute costs. It is the most flexible type offered. They can purchase a one-year All Upfront offering to maximize savings for the first year.

Reference: [What Are Savings Plans?](#)

Reference: [Purchasing Savings Plans](#)

✗ Purchase a one-year All Upfront EC2 Instance Savings Plan.

QUESTION 40

Your company has decided to migrate a SQL Server database to a newly-created AWS account. Which service can be used to migrate the database?

☐ DynamoDB

☐ AWS RDS

☐ Database Migration Service

☐ ElastiCache

✓ Database Migration Service

AWS Database Migration Service helps you migrate databases to AWS quickly and securely. The source database remains fully operational during the migration, minimizing downtime to applications that rely on the database. The AWS Database Migration Service can migrate your data to and from the most widely used commercial and open-source databases.

AWS Database Migration Service supports homogeneous migrations such as Oracle to Oracle, as well as heterogeneous migrations between different database platforms, such as Oracle or Microsoft SQL Server to Amazon Aurora. With AWS Database Migration Service, you can continuously replicate your data with high availability and consolidate databases into a petabyte-scale data warehouse by streaming data to Amazon Redshift and Amazon S3. Learn more about the supported source and target databases.

<https://aws.amazon.com/dms/>

QUESTION 41

Your team has provisioned Auto Scaling groups in a single Region. The Auto Scaling groups, at max capacity, would total 40 EC2 On-Demand Instances between them. However, you notice that the Auto Scaling groups will only scale out to a portion of that number of instances at any one time. What could be the problem?

- ☐ The associated load balancer can serve only 20 instances at one time.
- ☐ You can have only 20 instances per Region. This is a hard limit.
- ☐ There is a vCPU-based On-Demand Instance limit per Region.
- ☐ You can have only 20 instances per Availability Zone.

- ✓ There is a vCPU-based On-Demand Instance limit per Region.

Your AWS account has default quotas, formerly referred to as limits, for each AWS service. Unless otherwise noted, each quota is Region specific. You can request increases for some quotas, and other quotas cannot be increased. Remember that each EC2 instance can have a variance of the number of vCPUs, depending on its type and your configuration, so it's always wise to [calculate your vCPU needs](#) to make sure you are not going to hit quotas easily. Service Quotas is an AWS service that helps you manage your quotas for over 100 AWS services from one location. Along with looking up the quota values, you can also request a quota increase from the Service Quotas console.

Reference: [AWS Service Quotas](#) Reference: [Amazon EC2 Endpoints and Quotas](#)

QUESTION 42

A software company has developed a social gaming application that leverages EC2 web servers with Amazon DynamoDB to store player data, session history, and leaderboards for a huge number of concurrent users. The DynamoDB table has pre-configured read and write capacity units. Users have been reporting slowdown issues, and an analysis has revealed that the application requires response times in microseconds for optimal performance. What step can you take to enable this application to handle read-heavy or bursty workloads, while delivering the fastest possible response time for eventually consistent read operations?

- ☐ Configure Amazon SQS to queue requests that could be lost and improve the application response time.
- ☐ Add a load balancer in front of the EC2 web servers to decouple your application requests synchronously, improving performance for read-heavy and bursty workloads.
- ☐ Deploy Amazon CloudFront to your architecture, so you can cache common Amazon DynamoDB queries and reduce response time to microseconds.
- ☐ Implement in-memory acceleration with DynamoDB Accelerator (DAX).

- ✗ Configure Amazon SQS to queue requests that could be lost and improve the application response time.
- ✗ Add a load balancer in front of the EC2 web servers to decouple your application requests synchronously, improving performance for read-heavy and bursty workloads.
- ✗ Deploy Amazon CloudFront to your architecture, so you can cache common Amazon DynamoDB queries and reduce response time to microseconds.
- ✓ Implement in-memory acceleration with DynamoDB Accelerator (DAX).

DAX is a DynamoDB-compatible caching service that enables you to benefit from fast in-memory performance for demanding applications. As an in-memory cache, DAX reduces the response times of eventually consistent read workloads by an order of magnitude from single-digit milliseconds to microseconds. AWS also recommends DAX for read-heavy or bursty workloads, since it provides increased throughput and potential operational cost savings by reducing the need to overprovision read capacity units. AWS Documentation: [Use Cases for DAX](#).

QUESTION 43

You work for a large online education company that teaches IT using pre-recorded videos. They want to make their website available to the hearing impaired and need to find a way to convert their videos and audio into speech that they can then display as subtitles. Which AWS service should they use?

☐ Amazon Transcribe

☐ Amazon Translate

☐ Amazon Comprehend

☐ Amazon Rekognition

✓ Amazon Transcribe

Amazon Transcribe converts speech to text automatically. You can use this service to generate subtitles.

✗ Amazon Translate

✗ Amazon Comprehend

✗ Amazon Rekognition

QUESTION 44

A gaming company is designing several new games which focus heavily on player-game interaction. The player makes a certain move and the game has to react very quickly to change the environment based on that move and to present the next decision for the player in real-time. A tool is needed to continuously collect data about player-game interactions and feed the data into the gaming platform in real-time. Which AWS service can best meet this need?

☐ AWS Lambda

☐ Kinesis Data Analytics

☐ AWS IoT

☐ Kinesis Data Streams

- ✗ AWS Lambda
- ✗ Kinesis Data Analytics
- ✗ AWS IoT
- ✓ Kinesis Data Streams

Kinesis Data Streams can be used to continuously collect data about player-game interactions and feed the data into your gaming platform. With Kinesis Data Streams, you can design a game that provides engaging and dynamic experiences based on players' actions and behaviors. <https://aws.amazon.com/kinesis/data-streams/>

QUESTION 45

Currently, you are employed as a solutions architect for a large international shipping company. The company is undergoing an IT transformation and they want to create an immutable database, where they can track packages as they are sent around the world. They will need to track what boxes they go in, what trucks they are sent in, and what aircraft or sea containers they are shipped in. The database needs to be immutable and cryptographically verifiable, and they would like to leverage the AWS cloud to achieve this. What database technology would best suit this requirement?

☐ Amazon Quantum Ledger Database (QLDB)

☐ Neptune

☐ RDS

☐ Aurora

✓ Amazon Quantum Ledger Database (QLDB)

This is an immutable and cryptographically verifiable database and would be the best solution.

✗ Neptune

✗ RDS

✗ Aurora

QUESTION 46

You have recently migrated your small company to AWS and are looking for some general best practice guidance within the platform. Which AWS service can help you optimize your AWS environment by giving recommendations to reduce cost, increase performance, and improve security?

☐ AWS Trusted Advisor

☐ AWS Inspector

☐ AWS Optimizations

☐ AWS Organizations

✓ AWS Trusted Advisor

Correct: AWS Trusted Advisor is an online tool that provides you realtime guidance to help you provision your resources following AWS best practices. Trusted Advisor checks help optimize your AWS infrastructure, increase security and performance, reduce your overall costs, and monitor service limits. Whether establishing new workflows, developing applications, or as part of ongoing improvement, take advantage of the recommendations provided by Trusted Advisor on a regular basis to help keep your solutions provisioned optimally.

✗ AWS Inspector

✗ AWS Optimizations

✗ AWS Organizations

QUESTION 47

Jennifer is a cloud engineer for her application team. The application leverages several third-party SaaS vendors to complete their workflows within the application. Currently, the team uses numerous AWS Lambda functions for each SaaS vendor that run daily to connect to the configured vendor. The functions initiate a transfer of data files, ranging from one megabyte up to 80 gibibytes in size. These data files are stored in an Amazon S3 bucket and then referenced by the application itself. The data transfer routinely fails due to execution timeout limits in the Lambda functions, and the team wants to find a simpler and less error-prone way of transferring the required data. Which solution or AWS service could be the best fit for their solution?

- ☐ Amazon EC2 Auto Scaling Groups
- ☐ Amazon AppFlow
- ☐ Increase the Lambda function timeouts to one hour
- ☐ Amazon EKS with Auto Scaling

✗ Amazon EC2 Auto Scaling Groups

✓ Amazon AppFlow

AppFlow offers a fully managed service for easily automating the exchange of data between SaaS vendors and AWS services like Amazon S3. You can transfer up to 100 gibibytes per flow, and this avoids the Lambda function timeouts. Reference: [What is Amazon AppFlow? Tutorial: Transfer data between applications with Amazon AppFlow](#)

✗ Increase the Lambda function timeouts to one hour

✗ Amazon EKS with Auto Scaling

QUESTION 48

A financial institution has an application that produces huge amounts of actuary data, which is ultimately expected to be in the terabyte range. There is a need to run complex analytic queries against terabytes of structured data, using sophisticated query optimization, columnar storage on high-performance storage, and massively parallel query execution. Which service will best meet this requirement?

- ☐ Redshift
- ☐ RDS
- ☐ ElastiCache
- ☐ DynamoDB

✓ Redshift

Amazon Redshift is a fast, fully-managed cloud data warehouse that makes it simple and cost-effective to analyze all your data using standard SQL and your existing Business Intelligence (BI) tools. It enables you to run complex analytic queries against terabytes to petabytes of structured data, using sophisticated query optimization, columnar storage on high-performance storage, and massively parallel query execution. Most results come back in seconds. With Redshift, you can start small for just \$0.25 per hour with no commitments and scale out to petabytes of data for \$1,000 per terabyte per year, less than a tenth of the cost of traditional on-premises solutions. Amazon Redshift also includes Amazon Redshift Spectrum, allowing you to run SQL queries directly against exabytes of unstructured data in Amazon S3 data lakes. No loading or transformation is required, and you can use open data formats, including Avro, CSV, Grok, Amazon Ion, JSON, ORC, Parquet, RCFile, RegexSerDe, Sequence, Text, and TSV. Redshift Spectrum automatically scales query compute capacity based on the data retrieved, so queries against Amazon S3 run fast, regardless of data set size.

QUESTION 49

You are managing S3 buckets in your organization. This management of S3 extends to Amazon Glacier. For auditing purposes you would like to be informed if an object is restored to S3 from Glacier. What is the most efficient way you can do this?

- ☐ Create an SNS notification for any upload to S3
- ☐ Configure S3 notifications for restore operations from Glacier
- ☐ Create a Lambda function which is triggered by restoration of object from Glacier to S3
- ☐ Create a CloudWatch Event for uploads to S3

✗ Create an SNS notification for any upload to S3

✓ Configure S3 notifications for restore operations from Glacier

The Amazon S3 notification feature enables you to receive notifications when certain events happen in your bucket. To enable notifications, you must first add a notification configuration that identifies the events you want Amazon S3 to publish and the destinations where you want Amazon S3 to send the notifications. An S3 notification can be set up to notify you when objects are restored from Glacier to S3.

✗ Create a Lambda function which is triggered by restoration of object from Glacier to S3

✗ Create a CloudWatch Event for uploads to S3

QUESTION 50

Your team owns three separate AWS accounts: one for production, one for staging, and one for development. Recently, there has been a push from the CEO to begin breaking down costs to the most comprehensive, detailed level. In addition to this level of detail, the team needs to store daily comma-separated value (CSV) reports in Amazon S3 for ingestion into the company's internal analytics tooling.

What would be the most efficient solution for this scenario?

- ☐ Use AWS Cost and Usage Reports to generate reports with the required amount of detail. Set up Amazon EventBridge (Amazon CloudWatch Events) to trigger a rule to create and then export CSV reports daily to a centralized Amazon S3 bucket.
- ☐ Use AWS Budgets to alert and generate reports, and use AWS Lambda to pull data, generate CSV reports, and then push them to Amazon S3.
- ☐ Use AWS Cost and Usage Reports to generate reports, and have it export CSV reports daily to a centralized Amazon S3 bucket.
- ☐ Use AWS Budgets to alert and generate reports on current spend, and use AWS Fargate to pull data, generate CSV reports, and then push them to Amazon S3.

- ✗ Use AWS Cost and Usage Reports to generate reports with the required amount of detail. Set up Amazon EventBridge (Amazon CloudWatch Events) to trigger a rule to create and then export CSV reports daily to a centralized Amazon S3 bucket.
- ✗ Use AWS Budgets to alert and generate reports, and use AWS Lambda to pull data, generate CSV reports, and then push them to Amazon S3.
- ✓ Use AWS Cost and Usage Reports to generate reports, and have it export CSV reports daily to a centralized Amazon S3 bucket.

AWS Cost and Usage Reports offers the greatest amount of detail for spending reports. They can also be set up to automatically store updated reports in Amazon S3 every 24 hours.

- ✗ Use AWS Budgets to alert and generate reports on current spend, and use AWS Fargate to pull data, generate CSV reports, and then push them to Amazon S3.

QUESTION 51

You are a solutions architect working for an online gaming company. The company wants to integrate Amazon, Facebook, and Google authentication into the application so people can use their existing social media accounts to sign up to the application. Which AWS service facilitates this?

☐ Amazon Detective

☐ AWS Artifact

☐ Amazon Cognito

☐ AWS Trusted Advisor

✗ Amazon Detective

✗ AWS Artifact

✓ Amazon Cognito

Amazon Cognito provides authentication, authorization, and user management for your web and mobile apps in a single service without the need for custom code.

✗ AWS Trusted Advisor

QUESTION 52

You work for an advertising company that has a real-time bidding application. You are also using CloudFront on the front end to accommodate a worldwide user base. Your users begin complaining about response times and pauses in real-time bidding. What is the best service that can be used to reduce DynamoDB response times by an order of magnitude (milliseconds to microseconds)?

☐ ElastiCache

☐ DAX

☐ DynamoDB Auto Scaling

☐ CloudFront Edge Caches

✗ ElastiCache

✓ DAX

Correct. Amazon DynamoDB Accelerator (DAX) is a fully managed, highly available, in-memory cache that can reduce Amazon DynamoDB response times from milliseconds to microseconds, even at millions of requests per second. While DynamoDB offers consistent single-digit millisecond latency, DynamoDB with DAX takes performance to the next level with response times in microseconds for millions of requests per second for read-heavy workloads. With DAX, your applications remain fast and responsive, even when a popular event or news story drives unprecedented request volumes your way.

No tuning required. <https://aws.amazon.com/dynamodb/dax/>

✗ DynamoDB Auto Scaling

✗ CloudFront Edge Caches

QUESTION 53

An online retailer currently runs their application within AWS. Currently, everything is running on Amazon EC2 instances, including all application software. The application is well-written and completes order processes following a specific workflow logic order. The online retailer has begun to explore shifting their entire code base to AWS Lambda for each compute-based portion of the workflow, but they are not sure how best to interconnect the functions. There are three major requirements that need to be met. The first is that they need to implement a 20-minute wait period between certain functions in the application code and process. The second is they want to be able to conditionally handle a few different known scenarios that may occur during the order processing. The last requirement is to have an auditable workflow history. Which AWS service is the best fit for their workflow orchestration needs that has the least operational overhead and is the most cost-efficient?

☐ AWS Step Functions with AWS Lambda

☐ Amazon EKS with Amazon RDS

☐ AWS Lambda with Amazon SNS

☐ AWS Lambda with Amazon S3

✓ AWS Step Functions with AWS Lambda

Use AWS Step Functions to orchestrate Lambda functions for the computation. This service allows you to implement long-running workflows with wait periods and conditional catches. Reference: [What is AWS Step Functions? Wait Choice](#)

✗ Amazon EKS with Amazon RDS

✗ AWS Lambda with Amazon SNS

✗ AWS Lambda with Amazon S3

QUESTION 54

You are working in a large healthcare facility that uses EBS volumes on most of the EC2 instances. The CFO has approached you about some cost savings, and it has been decided that some of the EC2 instances and EBS volumes would be deleted. What step can be taken to preserve the data on the EBS volumes and ensure the data can be restored to a new EBS volume within minutes?

- ☐ Archive the data to Glacier.
- ☐ Store the data in CloudFormation user data.
- ☐ Take point-in-time snapshots of your Amazon EBS volumes.
- ☐ Move the data to Amazon S3.

- ✗ Archive the data to Glacier.
- ✗ Store the data in CloudFormation user data.
- ✓ Take point-in-time snapshots of your Amazon EBS volumes.

You can back up the data on your Amazon EBS volumes to Amazon S3 by taking point-in-time snapshots. Snapshots are incremental backups, which means that only the blocks on the device that have changed after your most recent snapshot are saved. This minimizes the time required to create the snapshot and saves on storage costs by not duplicating data. When you delete a snapshot, only the data unique to that snapshot is removed. Each snapshot contains all of the information that is needed to restore your data (from the moment when the snapshot was taken) to a new EBS volume. Reference: [Amazon EBS Snapshots](#)

- ✗ Move the data to Amazon S3.

QUESTION 55

Your application team stores critical data within a third-party SaaS cloud vendor. The data comes from an internal application that runs on Amazon ECS Fargate, which then stores the data within Amazon S3 in a proprietary format. Currently, AWS Lambda functions are triggered via Amazon S3 event notifications to trigger the transfer of data to a SaaS application. Due to resource and time limits, you are exploring other means of completing this workflow of transferring data from AWS to the SaaS solution. Which AWS service offers the most efficiency and has the least operational overhead?

- ☐ Amazon AppFlow
- ☐ AWS Step Function Fargate Capacity
- ☐ Amazon EKS
- ☐ Amazon EventBridge

✓ Amazon AppFlow

AppFlow offers a fully managed service for easily automating the bidirectional exchange of data to SaaS vendors from AWS services like Amazon S3. This helps avoid resource constraints.

Reference: [What Is AppFlow?](#) Reference: [Amazon AppFlow](#)

✗ AWS Step Function Fargate Capacity

✗ Amazon EKS

✗ Amazon EventBridge

QUESTION 56

You work for a large online education company that teaches IT using pre-recorded videos. You have converted their online videos into text to be accessible for the hearing impaired. They now want to convert the transcribed text into other languages using artificial intelligence and machine learning. Which AWS service should they use?

- ☐ Amazon Transcribe
- ☐ Amazon Rekognition
- ☐ Amazon Translate
- ☐ Amazon Comprehend

✗ Amazon Transcribe

✗ Amazon Rekognition

✓ Amazon Translate

Amazon Translate is a machine learning service that allows you to automate language translation.

✗ Amazon Comprehend

QUESTION 57

You work for a consulting company, which is currently undergoing both a financial and technical audit. You have been asked by the auditors to produce regular reports in regards to your PCI compliance. You need to produce this as fast and as efficiently as possible. Which AWS service should you consider using?

☐ Amazon Audit Automation

☐ Amazon Detective

☐ AWS Audit Manager

☐ AWS Security Hub

You work for a consulting company, which is currently undergoing both a financial and technical audit. You have been asked by the auditors to produce regular reports in regards to your PCI compliance. You need to produce this as fast and as efficiently as possible. Which AWS service should you consider using?

✗ Amazon Audit Automation

✗ Amazon Detective

✓ AWS Audit Manager

AWS Audit Manager is an automated service that produces reports specific to auditors for PCI compliance, GDPR, and more.

✗ AWS Security Hub

QUESTION 58

You work for an online retailer where any downtime at all can cause a significant loss of revenue. You have architected your application to be deployed on an Auto Scaling Group of EC2 instances behind a load balancer. You have configured and deployed these resources using a CloudFormation template. The Auto Scaling Group is configured with default settings and a simple CPU utilization scaling policy. You have also set up multiple Availability Zones for high availability. The load balancer does health checks against an HTML file generated by script. When you begin performing load testing on your application and notice in CloudWatch that the load balancer is not sending traffic to one of your EC2 instances. What could be the problem?

- ☐ The EC2 instance has failed EC2 status checks.
- ☐ You are load testing at a moderate traffic level and not all instances are needed.
- ☐ The EC2 instance has failed the load balancer health check.
- ☐ The instance has not been registered with CloudWatch.

- ✗ The EC2 instance has failed EC2 status checks.
- ✗ You are load testing at a moderate traffic level and not all instances are needed.

✓ The EC2 instance has failed the load balancer health check.

The load balancer will route the incoming requests only to the healthy instances. The EC2 instance may have passed status checks and be considered healthy to the Auto Scaling group, but the ELB may not use it if the ELB health check has not been met. The ELB health check has a default of 30 seconds between checks, and a default of 3 checks before making a decision. Therefore, the instance could be visually available but unused for at least 90 seconds before the GUI would show it as failed. In CloudWatch, where the issue was noticed, it would appear to be a healthy EC2 instance but with no traffic, which is what was observed.

<https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/elb-healthchecks.html>

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/as-add-elb-healthcheck.html>

- ✗ The instance has not been registered with CloudWatch.

QUESTION 59

An application team has decided to leverage AWS for their application infrastructure. The application performs proprietary, internal processes that other business applications utilize for their daily workloads. It is built with Apache Kafka to handle real-time streaming, which virtual machines running the application in docker containers consume the data from. The team wants to leverage services that provide less overhead but also cause the least amount of disruption to coding and deployments. Which combination of AWS services would best meet the requirements?

(Choose 2)

☐ AWS Lambda

☐ Amazon Kinesis Data Streams

☐ Amazon MSK

☐ Amazon ECS Fargate

☐ Amazon SNS

☐ Amazon MQ

✗ AWS Lambda

✗ Amazon Kinesis Data Streams

✓ Amazon MSK

This service is meant for applications that currently use or are going to use Apache Kafka for messaging. It allows for managing of control plane operations in AWS. Reference: [Welcome to the Amazon MSK Developer Guide](#)

✓ Amazon ECS Fargate

Fargate containers offer the least disruptive changes, while also minimizing the operational overhead of managing the compute services. Reference: [What is AWS Fargate?](#)

✗ Amazon SNS

✗ Amazon MQ

QUESTION 60

Your company has gotten back results from an audit. One of the mandates from the audit is that your application, which is hosted on EC2, must encrypt the data before writing this data to storage. It has been directed internally that you must have the ability to manage dedicated hardware security module instances to generate and store your encryption keys. Which service could you use to meet this requirement?

- ☐ AWS CloudHSM
- ☐ AWS KMS
- ☐ AWS Security Token Service
- ☐ Amazon EBS encryption

✓ AWS CloudHSM

The AWS CloudHSM service helps you meet corporate, contractual, and regulatory compliance requirements for data security by using dedicated Hardware Security Module (HSM) instances within the AWS cloud. A Hardware Security Module (HSM) provides secure key storage and cryptographic operations within a tamper-resistant hardware device. HSMs are designed to securely store cryptographic key material and use the key material without exposing it outside the cryptographic boundary of the hardware. You should use AWS CloudHSM when you need to manage the HSMs that generate and store your encryption keys. In AWS CloudHSM, you create and manage HSMs, including creating users and setting their permissions. You also create the symmetric keys and asymmetric key pairs that the HSM stores. AWS Documentation: [When to use AWS CloudHSM](#).

✗ AWS KMS

✗ AWS Security Token Service

✗ Amazon EBS encryption

QUESTION 61

Due to strict compliance requirements, your company cannot leverage AWS cloud for hosting their Kubernetes clusters, nor for managing the clusters. However, they do want to try to follow the established best practices and processes that the Amazon EKS service has implemented. How can your company achieve this while running entirely on-premises?

☐ Run the clusters on-premises using Amazon EKS Distro.

☐ Run Amazon ECS anywhere.

☐ Run Amazon EKS.

☐ This cannot be done.

✓ Run the clusters on-premises using Amazon EKS Distro.

Amazon EKS is based on the EKS Distro, which allows you to leverage the best practices and established processes on-premises that Amazon EKS uses in AWS. Reference:

[Amazon EKS Distro](#)

✗ Run Amazon ECS anywhere.

✗ Run Amazon EKS.

✗ This cannot be done.

QUESTION 62

A company needs to deploy EC2 instances to handle overnight batch processing. This includes media transcoding and some voice to text transcription. This is not high priority work, and it is OK if these batch runs get interrupted. What is the best EC2 instance purchasing option for this work?

☐ Spot

☐ Reserved

☐ On-Demand

☐ Dedicated Hosts

✓ Spot

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instance-purchasing-options.html>

Amazon EC2 provides the following purchasing options to enable you to optimize your costs based on your needs: On-Demand Instances – Pay, by the second, for the instances that you launch. Savings Plans – Reduce your Amazon EC2 costs by making a commitment to a consistent amount of usage, in USD per hour, for a term of 1 or 3 years. Reserved Instances – Reduce your Amazon EC2 costs by making a commitment to a consistent instance configuration, including instance type and Region, for a term of 1 or 3 years. Scheduled Instances – Purchase instances that are always available on the specified recurring schedule, for a one-year term. Spot Instances – Request unused EC2 instances, which can reduce your Amazon EC2 costs significantly. Dedicated Hosts – Pay for a physical host that is fully dedicated to running your instances, and bring your existing per-socket, per-core, or per-VM software licenses to reduce costs. Dedicated Instances – Pay, by the hour, for instances that run on single-tenant hardware. Capacity Reservations – Reserve capacity for your EC2 instances in a specific Availability Zone for any duration. <https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-spot-instances.html>

A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance is called a Spot price. The Spot price of each instance type in each Availability Zone is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Your Spot Instance runs whenever capacity is available and the maximum price per hour for your request exceeds the Spot price.

QUESTION 63

You are managing data storage for your company, and there are many EBS volumes. Your management team has given you some new requirements. Certain metrics on the EBS volumes need to be monitored, and the database team needs to be notified by email when certain metric thresholds are exceeded. Which AWS services can be configured to meet these requirements?

(Choose 2)

☐ SNS

☐ SQS

☐ SWF

☐ CloudWatch

☐ SES

✓ SNS

CloudWatch can be used to monitor the volume, and SNS can be used to send emails to the Ops team. Amazon SNS is for messaging-oriented applications, with multiple subscribers requesting and receiving "push" notifications of time-critical messages via a choice of transport protocols, including HTTP, Amazon SQS, and email.

✗ SQS

✗ SWF

✓ CloudWatch

CloudWatch can be used to monitor the volume, and SNS can be used to send emails to the Ops team. Amazon SNS is for messaging-oriented applications, with multiple subscribers requesting and receiving "push" notifications of time-critical messages via a choice of transport protocols, including HTTP, Amazon SQS, and email.

✗ SES

QUESTION 64

A small startup company has multiple departments with small teams representing each department. They have hired you to configure Identity and Access Management in their AWS account. The team expects to grow rapidly, and promote from within which could mean promoted team members switching over to a new team fairly often. How can you configure IAM to prepare for this type of growth?

- ☐ Create the user accounts, create a role for each department, create and attach an appropriate policy to each role, and place each user account into their department's role. When new team members are onboarded, create their account and put them in the appropriate role. If an existing team member changes departments, move their account to their new IAM group.
- ☐ Create the user accounts, create a group for each department, create and attach an appropriate policy to each group, and place each user account into their department's group. When new team members are onboarded, create their account and put them in the appropriate group. If an existing team member changes departments, move their account to their new IAM group.
- ☐ Create the user accounts, create a group for each department, create and attach an appropriate role to each group, and place each user account into their department's group. When new team members are onboarded, create their account and put them in the appropriate group. If an existing team member changes departments, move their account to their new IAM group.
- ☐ Create the user accounts, create a group for each department, create and attach an appropriate policy to each group, and place each user account into their department's group. When new team members are onboarded, create their account and put them in the appropriate group. If an existing team member changes departments, delete their account, create a new account and put the account in the appropriate group.

- ✓ Create the user accounts, create a group for each department, create and attach an appropriate policy to each group, and place each user account into their department's group. When new team members are onboarded, create their account and put them in the appropriate group. If an existing team member changes departments, move their account to their new IAM group.

An IAM group is a collection of IAM users. Groups let you specify permissions for multiple users, which can make it easier to manage the permissions for those users. For example, you could have a group called Admins and give that group the types of permissions that administrators typically need. Any user in that group automatically has the permissions that are assigned to the group. If a new user joins your organization and needs administrator privileges, you can assign the appropriate permissions by adding the user to that group. Similarly, if a person changes jobs in your organization, instead of editing that user's permissions, you can remove the user from the old groups and add the user to the appropriate new groups.

QUESTION 65

You have been put in charge of S3 buckets for your company. The buckets are separated based on the type of data they are holding and the level of security required for that data. You have several buckets that have data you want to safeguard from accidental deletion. Which configuration will meet this requirement?

- ☐ Signed URLs to all users to access the bucket.
- ☐ Configure cross-account access with an IAM Role prohibiting object deletion in the bucket.
- ☐ Enable versioning on the bucket and multi-factor authentication delete as well.
- ☐ Archive sensitive data to Amazon Glacier.

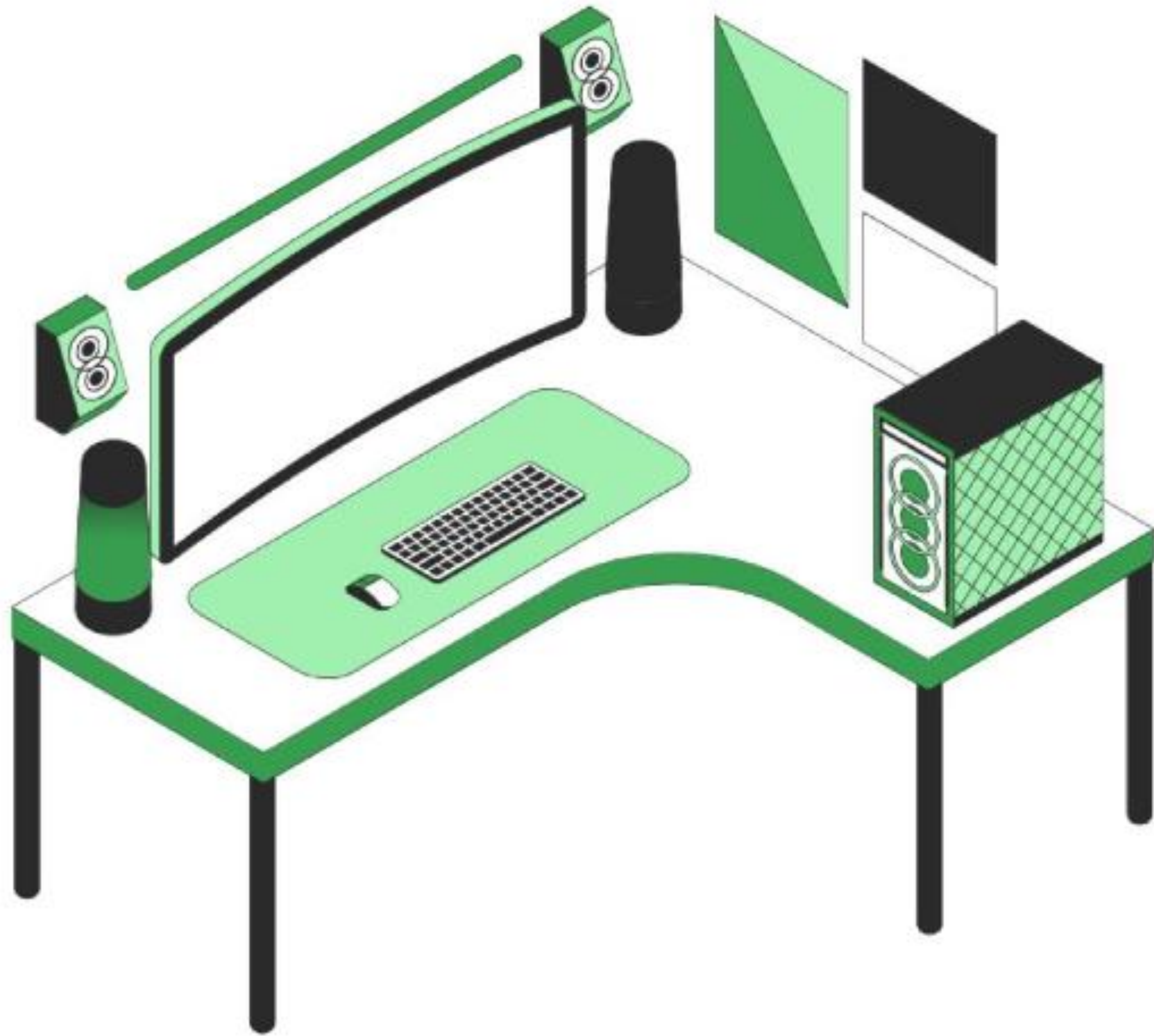
- ✗ Signed URLs to all users to access the bucket.
- ✗ Configure cross-account access with an IAM Role prohibiting object deletion in the bucket.
- ✓ Enable versioning on the bucket and multi-factor authentication delete as well.

Correct. Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. When you enable versioning for a bucket, if Amazon S3 receives multiple write requests for the same object simultaneously, it stores all of the objects. Key point: versioning is turned off by default. If a bucket's versioning configuration is MFA Delete-enabled, the bucket owner must include the x-amz-mfa request header in requests to permanently delete an object version or change the versioning state of the bucket. References:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/Versioning.html>

<https://docs.aws.amazon.com/AmazonS3/latest/dev/UsingMFADelete.html>

- ✗ Archive sensitive data to Amazon Glacier.



Do you
have any
questions?

Send it to us! We hope you learned
something new.